Households, Family, Fertility, and Time Allocation

Labor I — Week 6

Teaching deck draft

Central question and motivation

- Week 6 asks how households transform wage opportunities into realized hours, earnings, and career paths.
- Children and care needs reallocate time, not only tastes.
- Family policy matters for labor markets because it changes household production, bargaining, and dynamic career interruption.

Week 5 to Week 6 bridge

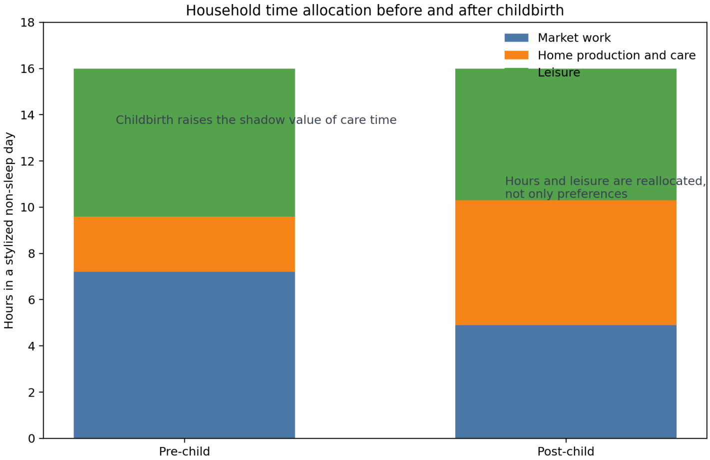
- Week 5 treated wages as market opportunities tied to skill and sorting.
- Week 6 asks why similar wage opportunities can produce different labor outcomes once partners, children, and care constraints enter.
- The relevant decision-making unit is often the household even when the data are person-level.

Household time-allocation problem

$$T_i = h_i + n_i + \ell_i$$

- Market work h_i, home production or care n_i, and leisure ℓ_i compete for the same time endowment.
- A birth shock changes the shadow value of n_i, not only the wage response along a standard labor-supply curve.
- Household-service substitutes and childcare access therefore matter for labor supply even when wages do not move.

Household time allocation before and after childbirth

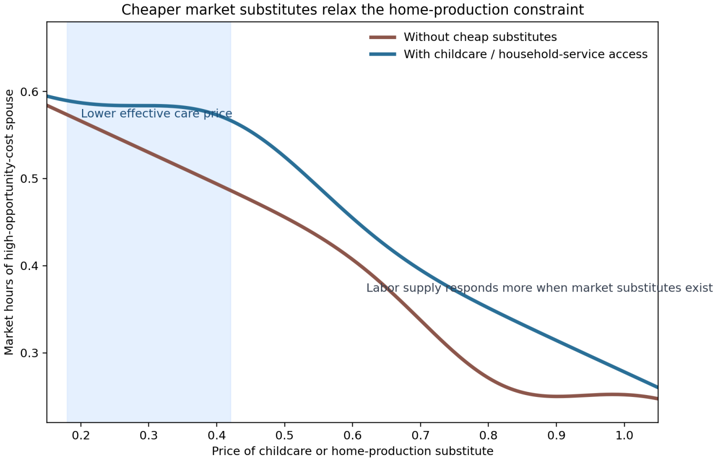


Household production and market substitutes

$$\max U(c_f, c_m, \ell_f, \ell_m, q) \quad \text{s.t.} \quad c_f + c_m + p_x x = w_f h_f + w_m h_m + y, \quad q = F(n_f, n_m, x)$$

- The household good q can be produced with parental time or market inputs x.
- Lower p_x relaxes the home-production constraint and can raise market hours.
- Gelbach and Cortés–Tessada are clean empirical versions of this comparative static.

Childcare and outsourcing as labor-supply shifters



Unitary versus collective household models

$$\max \lambda U_f(\cdot) + (1-\lambda) U_m(\cdot)$$

- Unitary models pool resources and act as a clean benchmark.
- Collective models keep efficiency but allow the sharing rule λ to respond to distribution factors.
- Policy incidence can therefore depend on who receives transfers, not only on total family income.

Household-model map

Bucket	Choice object	Variation or test	Good for	Caveat
Unitary	Pooled utility	Income pooling restrictions	Benchmark family labor supply	Too restrictive w recipient matters
Collective	Sharing rule λ	Distribution factors	Incidence and bargaining	Requires ric structure
Household production	Work, care, leisure, substitutes	Care prices and time prices	Care constraints and outsourcing	Home product hard to measure
Fertility IV	Extra child margin	IVF success or similar variation	Causal fertility effects	Local margin only
Child event study	Dynamic post-birth path	Event time around first birth	Persistent child penalties	Not an autom policy counterfact
Policy designs	Childcare, leave, transfers	Reform timing or eligibility	Policy incidence on labor outcomes	Multiple margins ten move togethe

Fertility and childbirth as labor-market shocks

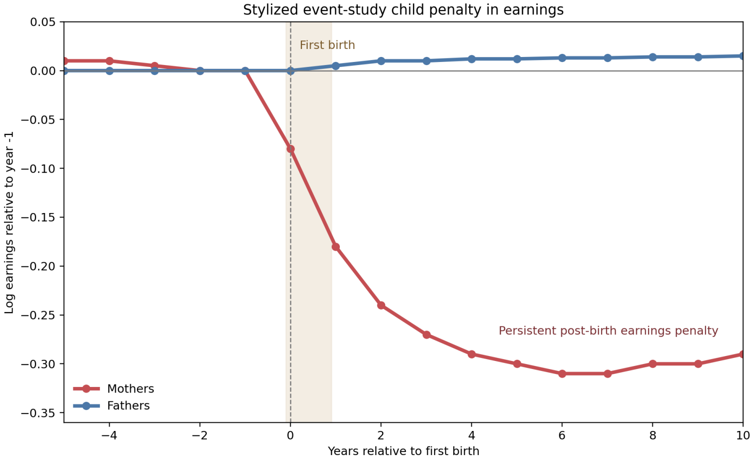
- Childbirth changes participation, hours, occupation choice, firm attachment, and later wage growth.
- Lundborg, Plug, and Rasmussen isolate a fertility margin through IVF treatment success.
- The resulting estimand is a local causal effect of an additional child, not a complete dynamic child-penalty path.

Child-penalty event-study object

$$Y_{it} = \sum_{k \neq -1} \beta_k \mathbf{1}\{t - b_i = k\} + \alpha_i + \gamma_t + \varepsilon_{it}$$

- The β_k coefficients trace the full path relative to the pre-birth period.
- Kleven, Landais, and Sogaard show that the divergence is persistent and tied to career dynamics, not only contemporaneous hours.
- This is why Week 6 is a bridge into inequality rather than only a family-policy week.

Stylized child-penalty event study



Policy-margin map

Policy or shock	Primary margin	Secondary margin	Outcomes	Week 6 use
Childbirth shock	Participation and hours	Wage growth, occupation, firm choice	Employment, earnings, wages	Fertility IV and penalties
Childcare access	Participation	Hours and re-entry	Employment, hours, childcare use	Childcare desig
Public school entry	Participation	Timing of work	Maternal labor supply	Gelbach-style th old
Parental leave	Short-run attachment	Long-run earnings path	Leave, retention, earnings	Family-policy e sions
Income to one spouse	Intra-household allocation	Labor supply and spending	Hours, expenditures	Pooling versus gaining tests
Cheaper services	Intensive margin	Leisure and housework	Hours and housework	Outsourcing cha

Bridge to Week 7 and Week 8

- Week 7: flexibility, schedule control, and remote work become household inputs once care constraints bind.
- Week 8: earnings inequality partly reflects fertility timing, care institutions, and household responses rather than only skill prices.
- Household structure is therefore central to both compensating differentials and modern inequality accounting.

Takeaways

- Family labor supply is jointly produced through wages, care needs, and household technology.
- Unitary and collective models imply different tests and different policy incidence.
- Fertility IV and child-event-study designs recover different but complementary objects.
- Week 6 is the bridge from worker-side labor supply to amenities and inequality.

Appendix: optional extension block

- Relative-income constraints and household norms in the spirit of Bertrand, Kamenica, and Pan.
- Long-run family-policy experimentation in the spirit of Kleven et al. (2024).
- Optional lab reflection: which policies change the home-production technology and which change bargaining weights?